MT 3 Review

- Tuesday (11/10)
- 8-9 PM ← Show up at 7:45 - 7:50
- Lectures 19-28 [See Below]
- WTHR 200 ← Main Floor NOT Balcony

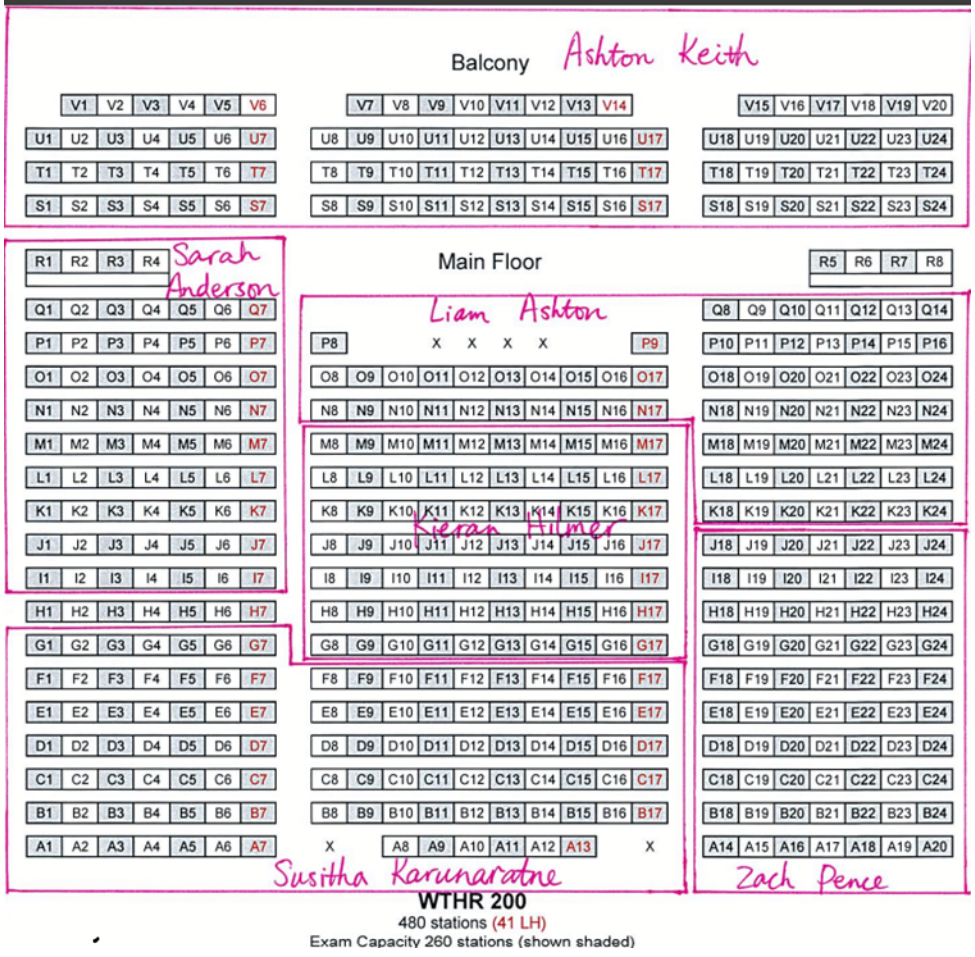
Additional Office Hours this week

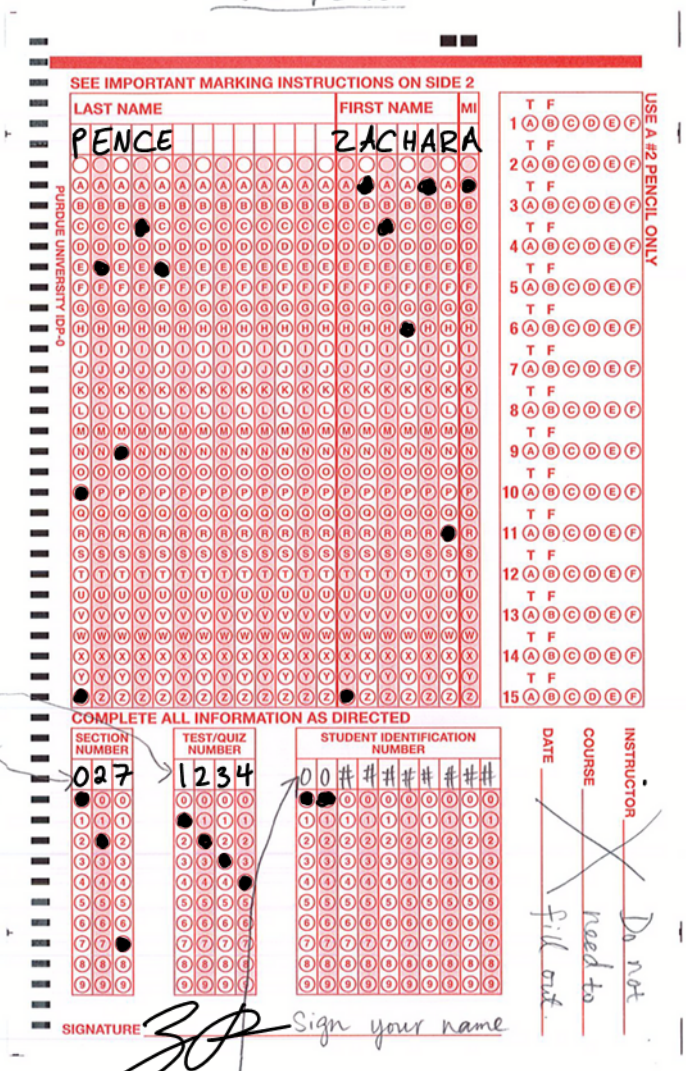
- Tuesday, 3:30 - 4:30 in my office (MATH 615)
- Usual office hours (Tues 2:30 - 3:30, MATH G175)

- No class Wednesday

Make Sure to Bring:

- Pencil / Erasers
- PUID
- Calculator TI 30XA Single Line Only





NOTE: These are just review problems of Lectures 19-28; these are not necessarily representative of the problems of the exam. The exam can (and most likely will) have different problems.

Topics of Exam 3:

- **Lecture 19:** Absolute Extrema on an Interval
- **Lecture 20:** Graphical Interpretation of Derivatives
- **Lecture 21:** Limits at Infinity
- **Lecture 22:** A Summary of Curve Sketching
- **Lecture 23-25:** Optimization
- **Lecture 26-27:** Antiderivatives and Indefinite Integration
- **Lecture 28:** Area and Riemann Sums

Problem 1. *Let M and m be the maximum and minimum of $f(x) = x^3 - 3x^2 + 3$ on the interval $[1, 4]$, respectively. What is the value of $M + m$?*

$x=1$, $x=4$, or any crit. num in $(1,4)$

$f(x) = x^3 - 3x^2 + 3$

$f'(x) = 3x^2 - 6x \overset{\text{set}}{=} 0$

$3x(x-2) = 0$

$x = 0, 2$

$x \in [1,4] \Rightarrow x = 2$

x	1	2	4
$f(x) = x^3 - 3x^2 + 3$	1	-1 (min)	19 (max)

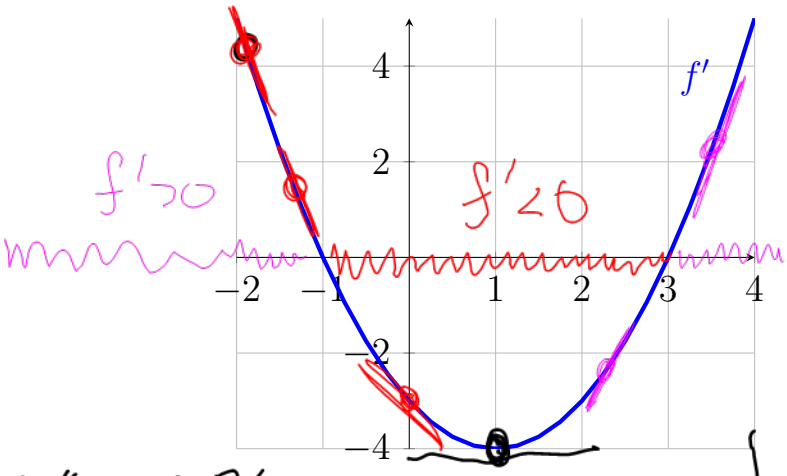
$M = 19$, $m = -1$, $M + m = 19 - 1 = 18$

Problem 2. *A graph of f' is given below.*

(a) Determine when f is increasing and when it is decreasing.

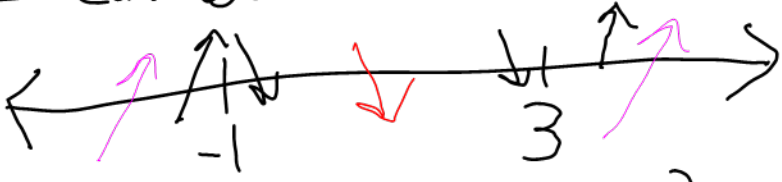
(b) Determine when f is concave up and when it is concave down.

(c) Locate the positions (x-coordinates) of any relative extrema and inflection points.



f'>0 f'<0

@ Conduct 1D Test



Inc: $(-\infty,-1) \cup (3,\infty)$

Dec: $(-1,3)$

Rel Max: at $x=-1$

Rel Min: at $x=3$

$f''=[f']'=0$ at $x=1$



f''<0 f''>0

↑ location of an IP

Problem 3. *Compute* $\lim_{x\to\infty} \frac{x^2+1}{2-x}$ *and* $\lim_{x\to-\infty} \frac{x^2+1}{2-x}$.

$$\lim_{x\to\infty} \frac{x^2+1}{2-x} = \lim_{x\to\infty} \frac{x^2}{-x} = \lim_{x\to\infty} -x = -\infty$$

$$\lim_{x\to-\infty} \frac{x^2+1}{2-x} = \lim_{x\to-\infty} -x = \infty$$

Problem 4. *Use the techniques learned in Lecture 22 to sketch a graph of the function* $y = \frac{x^2}{x+8}$.

$\frac{x^2}{x+8}$ ← deg is 2

← deg is 1

$x - 8$ ← slant asymptote

$$
\begin{array}{r}
x-8 \\
x+8 \overline{)\, x^2} \\
-(x^2+8x) \\
\hline
-8x \\
-(-8x-64) \\
\hline
64
\end{array}
$$

Problem 5. *If a rectangle has a fixed perimeter of* 40*, what is its maximum area?*



Objective: maximize area

$\max A_{(w,l)} = lw$

Given: $2w + 2l = 40$

$2l = 40 - 2w$

$l = 20 - w$

$\rightsquigarrow A(w) = (20 - w)w$

$A(w) = 20w - w^2$

$A'(w) = 20 - 2w \overset{\text{set}}{=} 0$

$20 = 2w$

$w = 10$

Need to compute $A(10)$:

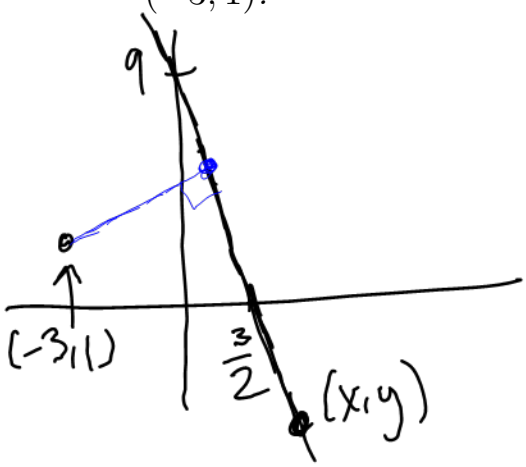
$A(w) = (20 - w)w$

$A(10) = (20 - 10)10 = 100$

Therefore, the max area is $100\ \text{un}^2$

The distance between (x,y) and

Problem 6. *Find the point on the line $6x+y=9$ that is closest to the point $(-3,1)$.*



Obj: Minimize distance

$$\min d_{(x,y)} = \sqrt{(x+3)^2+(y-1)^2}$$

It suffices to minimize

$$S(x,y) = [d(x,y)]^2 = (x+3)^2+(y-1)^2$$

Objective

Given: $6x+y=9 \Rightarrow y=-6x+9$

$$S(x) = (x+3)^2 + (-6x+9-1)^2$$

$$S(x) = (x+3)^2 + (-6x+8)^2$$

$$S(x) = x^2+6x+9 + 36x^2-96x+64$$

$$S(x) = 37x^2-90x+73$$

$$S'(x) = 74x-90 \overset{\text{set}}{=} 0$$

$$x = \frac{90}{74} = \frac{45}{37}$$

Need to find the y-coordinate:

$$y = -6x+9$$

$$y = -6\left(\frac{45}{37}\right)+9 = \frac{63}{37}$$

(x,y) (x_0,y_0)

$$\sqrt{(x-x_0)^2+(y-y_0)^2}$$

The point $\left(\frac{45}{37}, \frac{63}{37}\right)$ is closest to the point $(-3,1)$.

What is $S\left(\frac{45}{37}, \frac{63}{37}\right)$ ← Minimum squared distance

$$\sqrt{S\left(\frac{45}{37}, \frac{63}{37}\right)}$$

Problem 7. *A particle is moving in a straight line; its acceleration function is given by* $a(t) = \cos t + \sin t$ *for* $t \geq 0$. *Find the position of the particle at time* t *when the particle has an initial position of 0 and an initial velocity of 5.*

$$\begin{cases} a(t) = \cos t + \sin t \\ v(0) = 5 \\ s(0) = 0 \end{cases}$$

$a(t) = \cos t + \sin t$

$v(t) = \sin t - \cos t + C$

$v(0) = 5$ 5 should $v(0) = 0 - 1 + C$

$5 = -1 + C$

$C = 6$

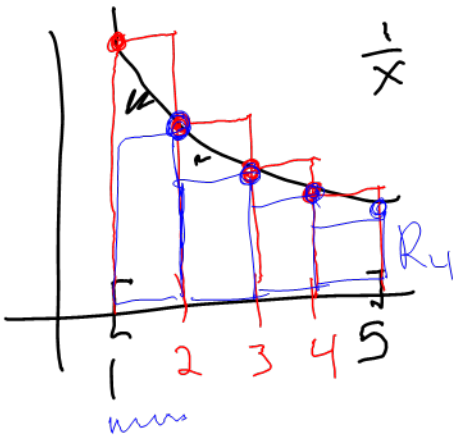
$v(t) = \sin t - \cos t + 6$

Problem 8. *We estimate the area underneath the graph of* $f(x) = \frac{1}{x}$ *on the interval* $[1, 5]$.

(a) *Compute the left Riemann sum with* 4 *rectangles. Is this an overestimate or an underestimate? Explain why (The exact area is* $\ln 5$*, but you don't need to know that to answer the question).*

(b) *Repeat (a) using the right Riemann sum.*

(c) *Set up (but do not compute) the left Riemann sum using* N *rectangles (where* N *is a positive integer).*



(a) Compute L_4

Width:

$$\Delta x = \frac{b-a}{N} = \frac{5-1}{4} = \frac{4}{4} = 1$$

$$L_N = \sum_{i=0}^{N-1} f(a + i\Delta x)\,\Delta x$$

$$L_4 = \sum_{i=0}^{3} f(1 + i(1))(1) = \sum_{i=0}^{3} f(1+i) = \sum_{i=0}^{3} \frac{1}{1+i}$$

$$= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{25}{12} \leftarrow \text{Over estimate}$$

(b) Compute R_4

$$R_4 = \sum_{i=1}^{4} \frac{1}{1+i} = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \frac{77}{60}$$

Under Est.

(c) Set Up L_N and R_N

$$\Delta x = \frac{b-a}{N} = \frac{5-1}{N} = \frac{4}{N}$$

$$L_N = \sum_{i=0}^{N-1} f(a + i\Delta x)\Delta x = \sum_{i=0}^{N-1} f\left(1 + i\left(\frac{4}{N}\right)\right)\left[\frac{4}{N}\right]$$

$$= \sum_{i=0}^{N-1} \frac{1}{1+\frac{4i}{N}} \left[\frac{4}{N}\right]$$

$$R_N = \sum_{i=1}^{N} f(a+i\Delta x)\Delta x = \sum_{i=1}^{N} \frac{1}{1+\frac{4i}{N}} \left[\frac{4}{N}\right]$$